Date: 30.10.2026
★ topper's notes

Linux Commands & RAID

File system · permissions · processes · networking · RAID levels: exam-ready in 8 sheets

Linux
Commands (★ HIGH YIELD)

RAID
Levels 0/1/5/6/10 (★ HIGH YIELD)

Permissions
chmod / chown (★)

Core Commands
ls -la · pwd · cd · mkdir · rm -rf
grep -r · find · tail -f · head
chmod NNN · chown · ps aux
df -h · free -m · ip a · wget
sudo · kill · top · ssh · scp

RAID Table
RAID 0: stripe — N×disk, 0 fault
RAID 1: mirror — disk_size, 1 fault
RAID 5: (N-1)×disk, 1 fault ★
RAID 6: (N-2)×disk, 2 fault
RAID 10: N/2×disk, 1/pair ★

Legend yellow=key term · •=must-memorise · =memory trick · =exam trap

High-yield list
① chmod 755 → rwxr-xr-x
② chmod 701 → rwx-----x (group=0!)
③ RAID 5 usable = (N-1)×disk
④ tail -f = real-time log
⑤ RAID ≠ backup
⑥ PHP is NOT a web server

Contents

1. Linux file & directory commands … p.2
2. Permissions: chmod & chown … p.3
3. Process, disk & network commands … p.4
4. RAID concepts & levels … p.5
5. RAID capacity formulas … p.6
6. RAID failure recovery … p.7
7. Night-before sheet + MCQ … p.8

Syllabus coverage
Chapter 18 OS syllabus covers Linux commands, file permissions, and RAID storage. This notebook covers all of it. Processes, scheduling, memory management, and deadlock have separate lessons.

Top 3 traps
① chmod 701 = rwx-----x (0=---)
② RAID 5 usable = (N-1)×disk
③ RAID ≠ backup

1 File & Directory Commands ★ Most tested

Linux is an open-source Unix-like OS kernel. It uses a **text-based CLI (terminal)** for administration. Commands follow: `command [options] [arguments]`. Case-sensitive.

ls -la all files+perms | **pwd** current dir | **mkdir / rmdir** create / remove | **rm -rf dir** delete recursive+force | **cp / mv** copy / move | **find . -name '*.txt'** recursive file search

Command	Flags	Purpose	Example
ls	-la	list all files (including hidden)	ls -la /home
pwd	—	print working directory	pwd → /home/user
cd	.. ~ /path	change directory; cd ~ → home	cd /var/log
mkdir	-p	create directory (and parents)	mkdir -p a/b/c
rm	-rf	remove recursively & force	rm -rf /tmp/old
cp / mv	-r	copy / move files	cp -r src/ dst/
cat	—	display file content	cat /etc/hosts
grep	-r -i -n	search text in files	grep -rn 'error' /var/log/
find	-name -type	recursive file search	find . -name '*.c'
wget / curl	-O	download from URL	wget https://example.com/f.tar
tar	-cvf / -xvf	archive / extract	tar -cvf archive.tar dir/

```
$ find . -name '*.[ch]' | xargs wc -l | tail -1
3456 total
→ counts all .c and .h lines recursively
```

```
$ ls -la /home/user
total 48 · -rw-r--r-- 1 user ... .bashrc
dotfile (.bashrc) = hidden, shown by -a
```

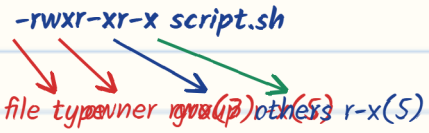
Memory trick **PCLMFRG**: pwd · cd · ls · mkdir · find · rm · grep — the 7 daily commands every sysadmin types.

Worked example Count lines in all .c and .h files: `find . -name '*.[ch]' | xargs wc -l | tail -1`

Exam trap `ls -la` shows hidden files (dotfiles starting with .). `ls -l` does NOT.

2 File Permissions: chmod & chown ★ Highest-yield

Permission model Every file has 3 user classes: **owner**, **group**, **others**. Each gets **r=4, w=2, x=1** bits. `chmod NNN file` — each digit = r×4+w×2+x×1.



Common chmod values
755 = rwxr-xr-x (executable/web dir)
644 = rw-r--r-- (regular files)
701 = rwx-----x (group=0, not --x!)
700 = rwx------ (private)
777 = rwxrwxrwx (dangerous!)
400 = r-------- (read-only)

Command	Purpose	Example
chmod 755 file	owner=rwx, group=r-x, others=r-x	standard for executables
chmod 644 file	owner=rw-, group=r--, others=r--	standard for data files
chmod 701 file	owner=rwx, group=---, others=--x	group gets NO bits (0=---)
chown user:group file	change owner and group	`chown www:www index.html`
chgrp group file	change group only	`chgrp dev project/`

Memory trick r=4, w=2, x=1 → add them: rwx=7, rw-=6, r-x=5, r--=4, --x=1, ---=0. Write the 3-digit octal left to right: owner · group · others.

Exam trap chmod **701**: middle digit is **0** = --- (no access for group), NOT --x. Many candidates write 701 as rwx--x--x — **WRONG**. 0 = ---, 1 = --x.

Worked example `chmod 755 script.sh`: 7=rwx(owner), 5=r-x(group), 5=r-x(others) → **-rwxr-xr-x**. Standard for web server directories.

Exam trap **chown** changes ownership. **chmod** changes permissions. They are separate operations — a common confusion in MCQs.

3 Process, Disk & Network Commands ★ PSU favourite

→ Process Management

Command	Purpose
ps aux	list all running processes
top	live CPU/memory usage
kill PID	terminate process by PID
kill -9 PID	SIGKILL — force terminate (uncatchable)
killall name	kill all processes by name
sudo command	run as root (audit trail)
su -	switch to root shell (not recommended)

→ Viewing File Content

Command	Purpose
tail -n 10 file	last 10 lines
tail -f file	follow live log (real-time!)
head -n 10 file	first 10 lines
less file	paginated view (q to quit)
wc -l file	count lines in file

→ Disk & Memory

Command	Purpose
df -h	disk space (human-readable)
du -sh /path	directory size
free -m	memory usage in MB
mount / umount	attach / detach filesystem

→ Network

Command	Purpose
ip a / ifconfig	show IP addresses
ping host	test connectivity
ssh user@host	remote login
scp file user@host:/path	secure file copy
wget URL	download file from URL
netstat / ss -tuln	show open ports

```
$ tail -f /var/log/syslog
Oct 30 10:05:12 server sshd[1234]: Accepted password for user
→ live stream — new lines appear as written. Ctrl+C to stop.
```

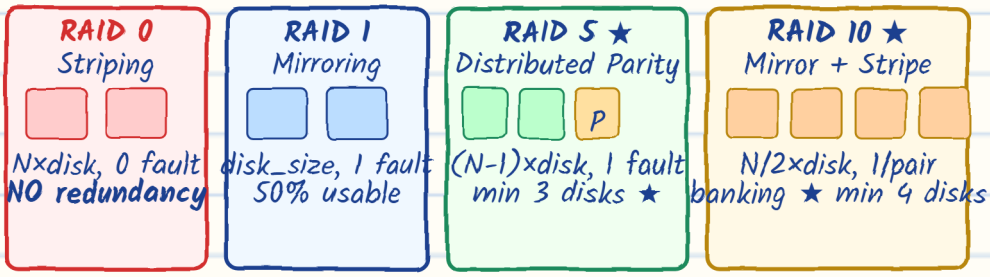
```
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       50G   20G   28G  42% /
```

Memory trick **tail -f** = "follow" = live log. **tail -n** = last N lines and exit. Exams ask which monitors a continuously updating log → tail -f.

Exam trap Root login is disabled by default for security. Attackers must guess username AND password. sudo logs every action with originating username — audit trail

4 RAID: Concepts & Levels ★ Bank/PSU favourite

RAID = Redundant Array of Independent Disks. Combines multiple physical disks into one logical unit. Goals: **performance** (striping) or **reliability** (mirroring/parity) or both.



Level	Technique	Min disks	Fault tolerance	Usable capacity	Best for
RAID 0	Striping	2	0	N × disk	Speed (video editing)
RAID 1	Mirroring	2	1 drive	disk_size	OS drives, critical data
RAID 5	Dist. parity	3	1 drive	(N−1) × disk	File servers, NAS
RAID 6	Double parity	4	2 drives	(N−2) × disk	Large arrays
RAID 10	Mirror+stripe	4 (even)	1 per pair	N/2 × disk	Banking databases ★

RAID 5 — 4 drives × 2TB



→ (4-1)×2 = **6 TB** usable
1 drive = parity (distributed across all 4)
1 drive failure → rebuild from parity ✓

Memory trick **RAID 5 usable = (N−1)×disk.** One disk "worth" of space goes to parity — distributed, but the math works as if one disk is consumed. Exams test this formula constantly.

Exam trap **RAID ≠ backup.** RAID protects against **hardware disk failure** only — not against ransomware, accidental deletion, or data corruption (those get mirrored too!).

5 RAID Capacity Formulas ★ Numerical favourite

Formula sheet Memorise these 5 formulas — they cover every RAID numerical in Bangladesh exams.

RAID	Usable capacity formula	4 × 2 TB example
RAID 0	N × disk_size	4 × 2 = 8 TB
RAID 1	disk_size (one copy)	2 TB (regardless of N)
RAID 5	(N − 1) × disk_size	(4−1) × 2 = 6 TB
RAID 6	(N − 2) × disk_size	(4−2) × 2 = 4 TB
RAID 10	(N / 2) × disk_size	(4/2) × 2 = 4 TB

Worked example 1 **BSCS AP 2026**: You have 4 drives of 2 TB each. Calculate usable storage for RAID 0, RAID 5, and RAID 10.

- RAID 0: 4 × 2 = **8 TB** (no redundancy — any failure loses all data)
- RAID 5: (4−1) × 2 = **6 TB** (1 drive for parity; tolerates 1 failure)
- RAID 10: (4/2) × 2 = **4 TB** (2 mirrored pairs striped together)

Worked example 2 **Combined Bank AME 2026**: 5 drives × 4 TB each in RAID 5. Why not 20 TB?

- Usable = (5−1) × 4 = **16 TB**
- 4 TB is consumed by distributed parity — one drive's equivalent across all 5
- Parity enables reconstruction if exactly ONE drive fails

Memory trick For **banking** questions: which RAID? → **RAID 10** (mirror pairs that stripe = best I/O performance + redundancy, no parity calculation delay). RAID 5 is acceptable for file/archive servers.

Exam trap RAID 10 needs an **even** number of disks (minimum 4). RAID 5 needs minimum 3. RAID 6 needs minimum 4. Confusing RAID 1 (min 2) with RAID 10 (min 4) is a common error.

Exam trap RAID 5 tolerates exactly **1** disk failure. Two simultaneous failures → data lost. Use RAID 6 for 2-drive tolerance.

6 RAID Failure Recovery & Drive Types

→ RAID failure recovery procedure

1. **Controller detects failure** → array continues in degraded mode (using remaining disks + parity)
2. **Replace failed drive** with a new drive of equal or greater capacity
3. **Controller rebuilds** (reconstructs) lost data from parity/mirror onto new drive automatically
4. During rebuild: array is **vulnerable** — a second failure in RAID 5 will lose all data

→ Server types

Type	Description
Tower	Standalone, like desktop PC
Rack	19" rack, measured in U units
Blade	Thin modules in shared chassis, highest density

→ Data centre tiers

Tier	Uptime
Tier 1	99.671%
Tier 2	99.741%
Tier 3	99.982% — NCC Bangladesh
Tier 4	99.995% (fault tolerant)

→ Linux root login

Root login is disabled by default for security:

① Attacker must guess both **username AND password**

② sudo logs every privileged action with username

③ Prevents accidental destructive root commands

→ Use `sudo command` for single commands

→ Use `su -` for full root shell (not recommended)

→ PUE formula

PUE = Total facility power / IT equipment power

PUE = 1.0 = perfect (impossible)

Industry average ≈ 1.5

Green data centres: PUE < 1.2

Lower PUE = more efficient

Memory trick Which RAID is best for a banking transaction server? → **RAID 10**: combines mirroring (full redundancy per pair) + striping (high throughput). No parity calculation penalty. Fast rebuild.

Exam trap RAID is **NOT a backup**. A RAID array protects against hardware disk failure. Ransomware, accidental deletion, and fire destroy all copies simultaneously. Always have a separate backup.

7 Night-Before Sheet & Practice MCQs

Fact sheet

- r=4, w=2, x=1; chmod NNN: owner·group·others
- chmod 755 → rwxr-xr-x; chmod 701 → rwx-----x
- ls -la shows hidden (dot) files
- tail -f = live/real-time log monitoring
- tail -n 10 = last 10 lines and exit
- find . -name '*.c' = recursive file search
- grep -r = recursive content search
- df -h = disk space; free -m = memory in MB
- RAID 0: N×disk, 0 fault (striping)
- RAID 1: disk_size, 1 fault (mirroring)
- RAID 5: (N-1)×disk, 1 fault (distributed parity, min 3)
- RAID 6: (N-2)×disk, 2 fault (double parity, min 4)
- RAID 10: N/2×disk, 1/pair (mirror+stripe, min 4)
- RAID ≠ backup; degraded mode on 1 failure
- PUE = total power / IT power (lower = better)
- Bangladesh NCC = Tier 3, Kaliakair, Gazipur

1. Which command monitors a continuously updating log file in real time?
a) tail -n 10 file b) head -f file **c) tail -f file** d) cat -r file

2. What is the permission string after `chmod 701 myfile`?
a) rwx-----x b) rwx--x--x c) rwxrwx--x d) rwx-wx--x

3. A RAID 5 array has 4 drives of 2 TB each. What is the usable storage?
a) 8 TB **b) 6 TB** c) 4 TB d) 2 TB

4. For a bank's core transaction server, which RAID is best?
a) RAID 0 b) RAID 5 c) RAID 1 **d) RAID 10**

5. Which Linux command shows disk space in human-readable format?
a) df -h b) du -h c) ls -h d) disk -h

6. RAID 0 uses 4 drives of 2 TB each. What is the total usable storage?
a) 8 TB b) 6 TB c) 4 TB d) 2 TB

7. Which RAID level provides fault tolerance through mirroring?
a) RAID 1 b) RAID 0 c) RAID 3 d) RAID 4

Answers: 1c 2a 3b 4d 5a 6a 7a

Written (5 marks): Explain RAID 0, 1, 5, and 10. Include usable capacity formula and fault tolerance for each. Which would you recommend for a Bangladesh bank's core transaction database and why?

Model answer:
RAID 0: N×disk, 0 fault — speed only, no redundancy.
RAID 1: disk_size, 1 fault — mirroring, 50% usable.
RAID 5: (N-1)×disk, 1 fault — distributed parity, min 3 disks, balance of space and safety.
RAID 10: N/2×disk, 1 per pair — mirror+stripe, min 4 disks.
Recommendation: **RAID 10** for banking — high I/O throughput (striping) + full redundancy (mirroring) + no parity calculation delay + fast rebuild. RAID 5 for less critical storage (reports, archives).